

EDUCATION	School of Computer Science, University of Sydney	Sydney, Australia
	<i>Ph.D., Computer Science</i>	2023 - 2026 (<i>expected</i>)
	• Advisors: Chang Xu & Daochang Liu • Research area: Uncertainty Calibration, Large (Vision) Language Models	
	School of Computer Science, University of Sydney	Sydney, Australia
	<i>M.Phil., Computer Science</i>	2021 - 2023
	• Advisors: Chang Xu & Lijun chang • Research area: Uncertainty Calibration, Neural Architecture Search	
	School of Computer Science, University of Sydney	Sydney, Australia
	<i>M.E., Data Science</i>	2020 - 2021
	Huazhong University of Science and Technology	Wuhan, China
	<i>B.E., Communication Engineering</i>	2013 - 2017
EXPERIENCES	• Received Science and Technology Scholarship	
	Research Intern Apple MLR, Cambridge UK	2026.05 - 2026.10
	• Implemented and scaled RL post-training pipelines for foundation models. • Profiled large-scale training infrastructure for throughput bottlenecks; reward hacking analysis and mitigation.	
	Research Intern Google Research, Sydney AU	2025.11 - 2026.02
	• Speculative decoding for efficient LLM inference, targeting decoding latency and serving throughput.	
	Teaching Assistant University of Sydney, Sydney	2022 - now
	• COMP5329 Deep Learning	Spring 2022, 2023, 2024, 2025
	• COMP5318 Machine Learning and Data Mining	Fall 2025
	• HTIN5005 Applied Healthcare Data Science	Fall 2022, 2023
	• OCMP5329 Deep Learning (Online)	Fall 2024, 2025
	• BUSS6002 Data Science in Business	Spring 2024, 2025 Fall 2024, 2025
	• QBUS5010 Intro to Dashboarding and Data Visualisation	Fall 2024, 2025
	Construction Supervisor Songjiang Construction Union, Shanghai	2017.07 - 2019.10
	IOS Developer Intern Trip.com, Shanghai	2015.06 - 2015.09
	Frontend Developer Intern Sunallies, Shanghai	2014.06 - 2014.09
TECHNICAL SKILLS	Languages & Frameworks: <i>Python, PyTorch</i>	
	Large-Scale Training: <i>Ray, Slurm, distributed RL post-training frameworks</i>	
	Inference & Serving: <i>vLLM, speculative decoding</i>	

AWARDS AND GRANTS	• Apple Scholars in AI/ML PhD Fellowship (20 selected worldwide, first Australian recipient), Apple 2026.04 • FFT Student Survey Award for Outstanding Achievement in BUSS6002, USYD Business School 2025.12 • AAAI 2026 Student Program Committee (3 selected worldwide), AAAI 2025.06 • International Tuition Fee Scholarship, USYD 2023.06 • Faculty of Engineering Research Support Scholarship, USYD 2023.05 • Excellent Student Cadre, HUST, 2015.09 • Science and Technology Scholarship, HUST 2014.06
ACADEMIC SERVICES	Conference Reviewer: <i>ICML, NeuIPS, ICLR, CVPR, ICCV, AAAI, IJCAI, ACMMM, AISTATS</i> Journal Reviewer: <i>T-MM, TMLR, DAMI, TPAMI, Machine Learning</i>
PUBLICATIONS	1. Haolan Guo, Linwei Tao, Haoyang Luo, Minjing Dong, Chang Xu. Sample Margin-Aware Recalibration of Temperature Scaling. <i>International Conference on Machine Learning, ICML 2026</i>. 2. Xiaoyang Li, Linwei Tao, Haohui Lu, Minjing Dong, Junbin Gao, Chang Xu. Calibrating Graph Neural Networks with Wavelet-Aware Temperature Scaling. <i>International Conference on Learning Representations, ICLR 2026</i>. 3. Lei Pan, Zhipeng Lu, Yuan Zheng, Chongyao Yan, Haitao Wen, Linwei Tao, Chang Xu. Task-adaptive continual learning of vision language models via prototype routing and prompt. <i>Neurocomputing</i>, 2026. 4. Haoyang Luo, Linwei Tao, Minjing Dong, Chang Xu. Beyond One-Hot Labels: Semantic Mixing for Model Calibration. <i>International Conference on Machine Learning, ICML 2025</i>. 5. Jinxu Lin, Linwei Tao, Minjing Dong, Chang Xu. Uncertainty Weighted Gradients for Model Calibration. <i>The IEEE / CVF Computer Vision and Pattern Recognition Conference, CVPR 2025</i>. 6. Jinxu Lin, Linwei Tao, Minjing Dong, Chang Xu. Diffusion Attribution Score: Evaluating Training Data Influence in Diffusion Model. <i>International Conference on Learning Representations, ICLR 2025</i>. (Spotlight, 5.1% of 11,500 submissions) 7. Linwei Tao, Minjing Dong, Chang Xu. Feature Clipping for Uncertainty Calibration. <i>AAAI Conference on Artificial Intelligence, AAAI 2025</i>. 8. Linwei Tao, Yi-Fan Yeh, Minjing Dong, Tao Huang, Philip Torr, Chang Xu. Revisiting Uncertainty Estimation and Calibration of Large Language Models. <i>Conference on Neural Information Processing Systems, NeurIPS 2025 Scaling Environments for Agents (SEA) Workshop</i> 9. Yingqing Yuan, Linwei Tao, Haohui Lu, Matloob Khushi, Imran Razzak, Mark Dras, Jian Yang, Usman Naseem. KG-UQ: Knowledge Graph-Based Uncertainty Quantification for Long Text in Large Language Models. <i>International World Wide Web Conference, WWW 2025 Workshop on Sustainable AI for the Future Web</i> 10. Lei Pan, Wuyang Luan, Yuan Zheng, Junhui Li, Linwei Tao, Chang Xu. GraphFusion: Integrating Multi-Level Semantic Information with Graph Computing for Enhanced 3D Instance Segmentation. <i>Neurocomputing</i>, 2024. 11. Linwei Tao, Younan Zhu, Haolan Guo, Minjing Dong, Chang Xu. A Benchmark Study on Calibration. <i>International Conference on Learning Representations, ICLR 2024</i>. 12. Linwei Tao, Minjing Dong, Chang Xu. Dual Focal Loss for Calibration. <i>International Conference on Machine Learning, ICML 2023</i>. 13. Linwei Tao, Minjing Dong, Daochang Liu, Changming Sun, Chang Xu. Calibrating a Deep Neural Network with Its Predecessors. <i>International Joint Conference on Artificial Intelligence, IJCAI 2023</i>.

1. **Linwei Tao**, Yi-Fan Yeh, Bo Kai, Minjing Dong, Tao Huang, Tom A. Lamb, Jialin Yu, Philip H.S. Torr, Chang Xu. Can Large Language Models Express Uncertainty Like Human?
2. Younan Zhu, **Linwei Tao**, Minjing Dong, Chang Xu. Attention Calibration for Reducing Hallucination in Large Vision-Language Models.
3. Yunke Wang, **Linwei Tao**, Bo Du, Yutian Lin, Chang Xu. Visual Imitation Learning with Calibrated Contrastive Representation.
4. **Linwei Tao**, Haoyang Luo, Minjing Dong, Chang Xu. Confidence Calibration under Ambiguous Ground Truth. *arXiv:2603.22879*, 2026.
5. **Linwei Tao**, Haolan Guo, Minjing Dong, Chang Xu. Consistency Calibration: Improving Uncertainty Calibration via Consistency among Perturbed Neighbors.
6. Rui Xu, Yunke Wang, **Linwei Tao**, Wenjie Xuan, Yong Luo. Diversifying Personalized Research Ideation against AI-Induced Homogenization. *arXiv:2607.28087*, 2026.